STUDY ONLY: one of four independent sensor / servo channels.
External RUN latch / command gating / PI tracking REQUIRED; not implemented here.

#FLG1
PWR_FLAG
#FLG2
PWR_FLAG
#FLG3
PWR_FLAG

ISO_DC 1
HGND 2
ISO_DC 3
HGND 4
ISO_LDO 5
INP 6
INN 7
HGND 8
DCDC_OUT
DCDC_HGND
HLDO_IN
NC
HLDO_OUT
INP
INN
HGND
DCDC_IN 16
DCDC_GND 15
DIAG 14
LDO_OUT 13
VDD 12
OUTP 11
OUTN 10
GND 9
U1
AMC3302DWE
LV_DC
PGND
DIAG
LV_DC
AUX5
OUTP
OUTN
PGND

U2A
DIFF_POS 3
DIFF_NEG 2
VMON_RAW 1
OPA4192PW

U2B
VMON 5
PI_NEG 6
PI_DRIVE 7
OPA4192PW

U2C
VREF_IN 10
VREF 9
VREF 8
OPA4192PW

PHASE_OUT
SH1
2m 0.1% / >=3W / 4-terminal
VOUT_COMMON

J1
PHASE_OUT 1
VOUT_COMMON 2
PHASE OUTPUT / COMMON OUTPUT

J2
AUX5 1
PGND 2
VREF_IN 3
ICMD 4
AUX5 / PGND / REF1V25 / ICMD

J3
ITH_LOCAL 1
RUN_LOCAL 2
DIAG 3
PGND 4
ITH / RUN / DIAG / PGND

CLAMP_DRAIN
Q1
CLAMP_GATE 1
CSD13381F4 / LANDPATTERN TBD
PGND

U2D
PGND 12
SPARE 13
SPARE 14
OPA4192PW

AUX5
U2E
V+ 4
V- 11
OPA4192PW
PGND

Positive current -> OUTP - OUTN rises -> VMON rises -> PI gate rises -> Q1 sinks local ITH.
HGND follows VOUT_COMMON near 100 V. PGND is floating secondary return. Never join HGND and PGND.

R1 10 0.1% PHASE_OUT INP
R2 10 0.1% VOUT_COMMON INN
R3 0 / EMI FB TBD VOUT_COMMON HGND
R4 47k AUX5 DIAG
R5 10k 0.01% ratio OUTP DIFF_POS
R6 10k 0.01% ratio VREF DIFF_POS
R7 10k 0.01% ratio OUTN DIFF_NEG
R8 10k 0.01% ratio VMON_RAW DIFF_NEG
R9 1k VMON_RAW VMON
R10 47k ICMD PI_NEG
R11 22k PI_DRIVE PI_CAP
R12 100 PI_DRIVE CLAMP_GATE
R13 100k CLAMP_GATE PGND
R14 470 ITH_LOCAL CLAMP_DRAIN
R15 100k ICMD PGND
R16 10k RUN_LOCAL PGND

C1 100nF 50V ISO_DC HGND
C2 1uF 25V ISO_DC HGND
C3 1nF 50V ISO_DC HGND
C4 100nF 50V LV_DC PGND
C5 100nF 50V ISO_LDO HGND
C6 1nF 50V ISO_LDO HGND
C7 1uF 25V AUX5 PGND
C8 1nF 50V AUX5 PGND
C9 10nF 50V C0G INP INN
C10 100nF 50V VMON PGND
C11 22nF 50V C0G PI_CAP PI_NEG
C12 100nF 50V AUX5 PGND

TP1 PHASE_OUT
TP2 VOUT_COMMON
TP3 HGND
TP4 AUX5
TP5 DIAG
TP6 OUTP
TP7 OUTN
TP8 VMON
TP9 ICMD
TP10 PI_DRIVE
TP11 ITH_LOCAL
TP12 RUN_LOCAL

Anti-windup, RUN guard and fault latch missing
Independent ITH / SS / V5 required for each phase
No PCB; no integrated four-phase hardware

Sheet: /
File: phase_sensor_study.kicad_sch

Title: STUDY ONLY: ONE PHASE SENSOR / ITH SINK

Size: A3 | Date: | Rev:
KiCad E.D.A. 9.0.2 | Id: 1/1